# GlyphJS — Complete Feature Showcase

GlyphJS is a Markdown-to-interactive-UI rendering engine. You write ordinary Markdown enriched with `ui:` fenced code blocks, and the pipeline compiles everything into a polished, interactive document — locally, with no server, no build step, and no cloud dependency.

This document demonstrates two things in sequence: first, that GlyphJS renders all standard Markdown faithfully; second, that it extends Markdown with 28 interactive components covering data visualization, diagrams, content layout, and interactive widgets.

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## Markdown Reference

A systematic index of every CommonMark element. Use this section to see exactly how the active theme colors, spaces, and contrasts each element.

## All heading levels

# H1 — Primary accent color

## H2 — Secondary palette color

### H3 — Heading with accent left bar

#### H4 — Heading text color, no decoration

##### H5 — MUTED · UPPERCASE · TRACKED

###### H6 — SMALLEST · MUTED · UPPERCASE

### Inline typography

Plain body text flows at normal weight. Within a sentence: **bold text** picks up the heading color, *italic text* stays in body color but slants, ***bold italic*** combines both, ~~strikethrough~~ draws a line through, and `inline code` switches to the monospace code palette with its own background tint.

You can nest freely: **bold containing `monospace code`**, *italic containing **nested bold***. Long italic runs look like this: *this entire sentence is set in italics to show the italic rendering across a fuller run of theme-colored body text — note the weight, slant angle, and color.*

Hard line breaks are two trailing spaces:
This line starts on a new line but belongs to the same paragraph block.

Escaped characters render literally: *not italic* `not code` [not a link] # not a heading.

Special HTML entities: & < > © — …

## Unordered lists

- First top-level item
- Second item with **bold**, *italic*, and `code` inline
  - Nested item at depth 2 — note the accent-colored marker
  - Another nested item
    - Depth 3 — deeply nested, still themed
    - A second item at depth 3
  - Back to depth 2
- Third top-level item

## Ordered lists

1. First step
2. Second step with a longer description that intentionally wraps to demonstrate how multi-line list items look in the current theme — does the leading align cleanly with the text?
3. Third step
   1. Sub-step one
   2. Sub-step two
      1. Sub-sub-step — three levels deep
4. Fourth step, back at the top level

## Mixed nesting

- Category A
  1. First ordered item inside an unordered list
  2. Second ordered item
- Category B
  - Unordered sub-item
    1. Ordered inside unordered inside unordered
- Category C, no nesting

## Blockquotes

> A simple single-paragraph blockquote. The background is `--glyph-surface`, the left border is `--glyph-accent`, and the text uses `--glyph-text-muted`.

> A multi-paragraph blockquote.The second paragraph continues here. **Bold** and *italic* and `code` work normally within blockquotes.— An attribution line in the same blockquote.

Nested blockquotes go one level deeper:

> Outer level of a nested quote. This sets up the context.

## Code blocks

An inline span: `const ir = await compile(markdown)` — monospace, code background, code text color.

A TypeScript fenced block:

```
interface ThemeFileData {
  base: 'light' | 'dark';
  overrides: Record<string, string>;
}

function resolveThemeVars(
  base: ThemeFileData['base'],
  overrides: ThemeFileData['overrides'],
): Record<string, string> {
  const baseVars = base === 'dark' ? DARK_THEME_VARS : LIGHT_THEME_VARS;
  return { ...baseVars, ...overrides };
}
```

A shell script:

```
# Export to every theme in one pass
for theme in themes/*.yml; do
  name=$(basename "$theme" .yml)
  glyphjs export doc.md \
    --format pdf \
    --theme-file "$theme" \
    --continuous \
    --padding "2rem 2.5rem" \
    -o "demo/demo-${name}.pdf"
done
```

A plain fenced block with no language tag:

```
This is unformatted preformatted text.
Monospace font, code background, no syntax highlighting.
Useful for output samples, ASCII diagrams, or terminal dumps.
```

## Tables

A standard table with a header row, body rows, and alternating stripe:

| Variable | Role | Dark default | Light default | | --- | --- | --- | --- | | `--glyph-bg` | Page background | `#1a1a1a` | `#ffffff` | | `--glyph-accent` | Primary accent | theme-defined | theme-defined | | `--glyph-surface` | Raised surface | `#242424` | `#f5f5f5` | | `--glyph-heading` | Heading text | `#f0f0f0` | `#1a1a1a` | | `--glyph-text-muted` | Secondary text | `#888` | `#666` | | `--glyph-border` | Subtle dividers | `#333` | `#e0e0e0` | | `--glyph-code-bg` | Code background | `#111` | `#f0f0f0` | | `--glyph-code-text` | Code text | `#e0e0e0` | `#333` |

## Horizontal rules

Thematic breaks created with `---`:

---

And with `***`:

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## Links

A regular link to the GlyphJS docs uses `--glyph-link` color with a soft underline tinted from `--glyph-accent-muted`. On hover it shifts to `--glyph-link-hover`.

An inline link with a title carries a tooltip title attribute.

Autolinks: https://glyphjs.dev and bare reference links are rendered the same way.

## Images

Images are max-width constrained and corner-rounded. A missing image shows the alt text:

The GlyphJS pipeline: Markdown → Parser → Compiler → IR → Runtime → DOM

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# Standard Markdown

Everything in this section is plain CommonMark. GlyphJS passes it through unchanged, rendered by whatever host environment you're using — a browser, Obsidian, a PDF export, or a static site.

## Text formatting

Paragraphs are separated by blank lines. Within a paragraph you can use **bold text** for emphasis, *italic text* for softer stress, `inline code` for identifiers and values, and ~~strikethrough~~ when you need to show something has been removed.

You can combine these freely: a `typed constant` or an *italicized phrase with a **nested bold** inside it*. GlyphJS does not restrict how Markdown inline syntax nests.

## Lists

Unordered lists work with `-`, `*`, or `+` markers:

- Parser — transforms `ui:*` fenced blocks into AST nodes via a remark plugin
- Compiler — walks the AST and produces a typed JSON Intermediate Representation (IR)
- Runtime — a React rendering engine that takes the IR and produces interactive DOM
- Components — 28 built-in components, each with a Zod schema and a React renderer

Ordered lists use numeric prefixes:

1. Write a Markdown file with embedded `ui:` code blocks
2. Run `glyphjs export document.md --format pdf -o output.pdf`
3. Open the PDF — components are fully rendered, no JavaScript required

Lists can nest up to any depth:

- `@glyphjs/types` — shared TypeScript interfaces
  - `Block` — the fundamental unit of the IR
  - `GlyphIR` — the complete document representation
- `@glyphjs/schemas` — Zod schemas for all 28 components
  - Used for validation at parse time
  - Used for JSON Schema generation via `glyphjs schemas`

## Blockquotes

> **Design principle:** A document format should compose with its host environment, not fight it. GlyphJS components are defined in plain YAML inside standard fenced code blocks — any Markdown editor that doesn't know GlyphJS shows the YAML source; any environment that does renders the component.

Blockquotes can hold any block-level content, including nested quotes:

> The pipeline is stateless and deterministic.

## Code blocks

Fenced code blocks with language identifiers get syntax highlighting in most renderers:

```
import { compile } from '@glyphjs/compiler';

const ir = await compile(`
# My Document

\`\`\`ui:callout
type: tip
title: Hello world
content: GlyphJS compiled this.
\`\`\`
`);

console.log(ir.blocks[0].type); // "callout"
```

```
# A GlyphJS theme file — only override what you need.
# The base system fills in the rest.
base: dark
variables:
  --glyph-bg: '#1e1e2e'
  --glyph-accent: '#cba6f7'
  --glyph-heading: '#cdd6f4'
```

## Tables

Markdown tables are rendered as styled HTML tables:

| Package | Role | Key export |
|---|---|---|
| `@glyphjs/types` | Type definitions | `Block`, `GlyphIR` |
| `@glyphjs/schemas` | Validation | `componentSchemas`, `getJsonSchema` |
| `@glyphjs/compiler` | Compilation | `compile()` |
| `@glyphjs/runtime` | React rendering | `createGlyphRuntime()` |
| `@glyphjs/cli` | CLI tooling | `glyphjs` binary |

### Links and images

The GlyphJS documentation covers the full component reference, theming guide, and CLI usage. The source is at github.com/VledicFranco/glyphjs.

### Horizontal rules

A thematic break separates sections with visual weight:

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Everything above is standard Markdown. Everything below is rendered by GlyphJS.

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## GlyphJS Components

The following components are rendered by the GlyphJS runtime. Each `ui:` block is a YAML object validated against its Zod schema, compiled to IR, and mounted as a React component.

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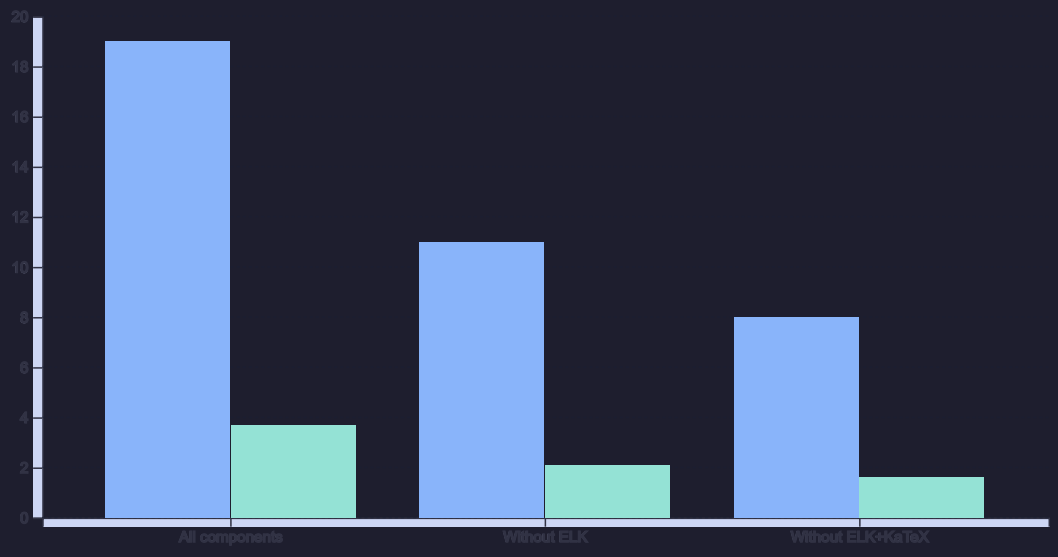
## Callout

> 💡 **Plugin is working!**
> GlyphJS renders natively in Obsidian with full theme sync.

## Table

| Component | Status | Phase |
|---|---|---|
| Callout | ✓ Rendering | 1 |
| Table | ✓ Rendering | 1 |
| Chart | ✓ Rendering | 1 |
| Live Preview | Pending | 2 |

## Chart




## KPI

Components
**28** total

Bundle (gzip)
**3.7** MB

Phase
**1** of 3

## Steps

### Phase 1 — Reading view

Bundle all packages. Register MarkdownPostProcessor. Mount React roots via MarkdownRenderChild. Ship with theme sync.

### Phase 2 — Live preview

CM6 StateField with widget decorations. Cursor-aware block hiding. Debounced compile().

### Phase 3 — Deep integration

Full theme mapping. Settings panel. Export-to-PDF command. Mobile support.

## Timeline

TBD

**Phase 2 — Live preview**

CM6 StateField for in-editor component preview

Mar 1, 2026

**Reading view working**

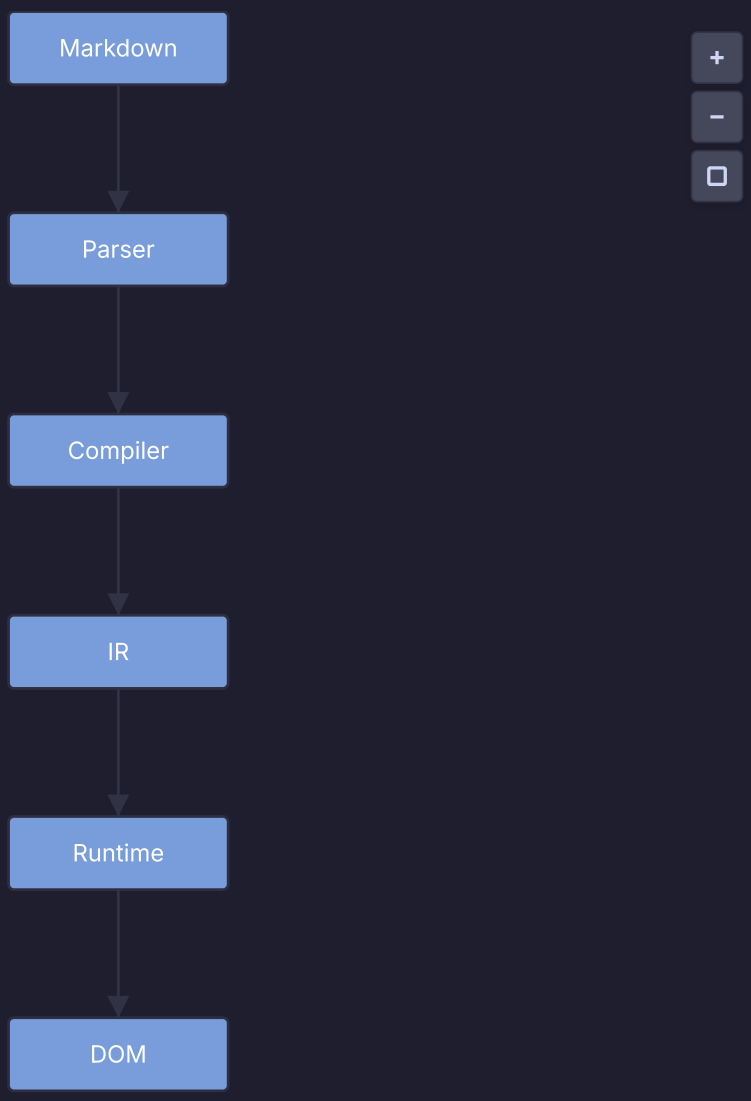
All 28 components rendering with Obsidian theme sync confirmed

Feb 28, 2026

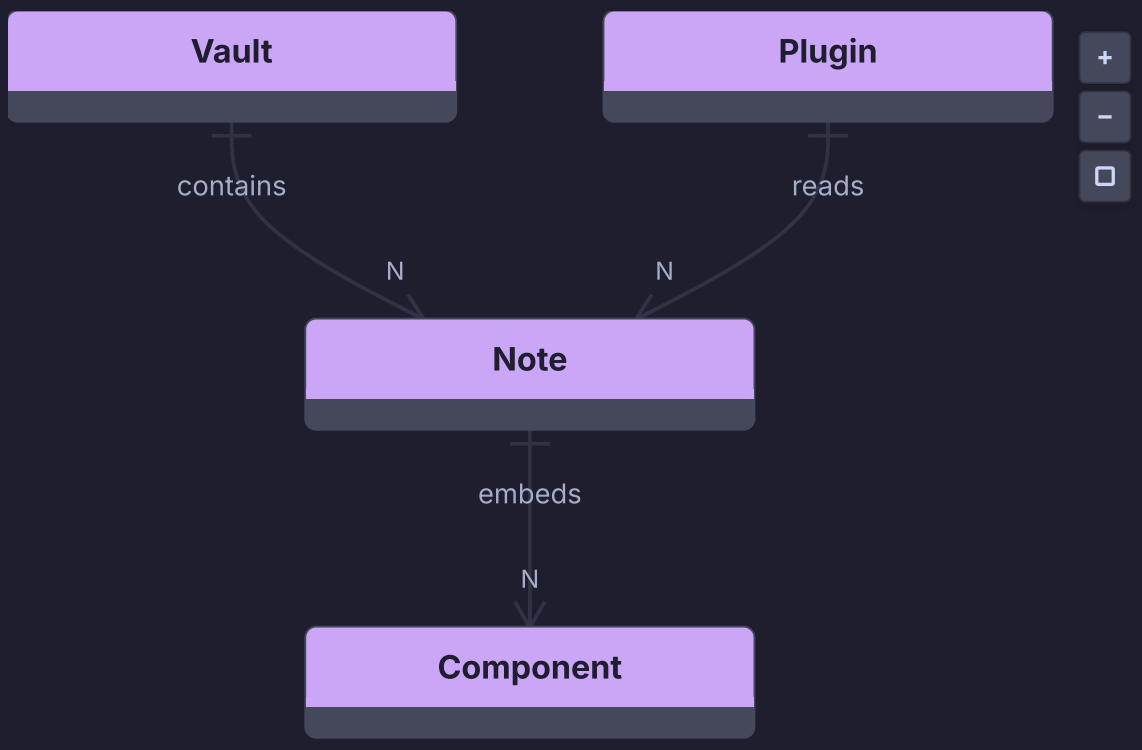
**POC started**

Created packages/obsidian-plugin in the glyphjs monorepo

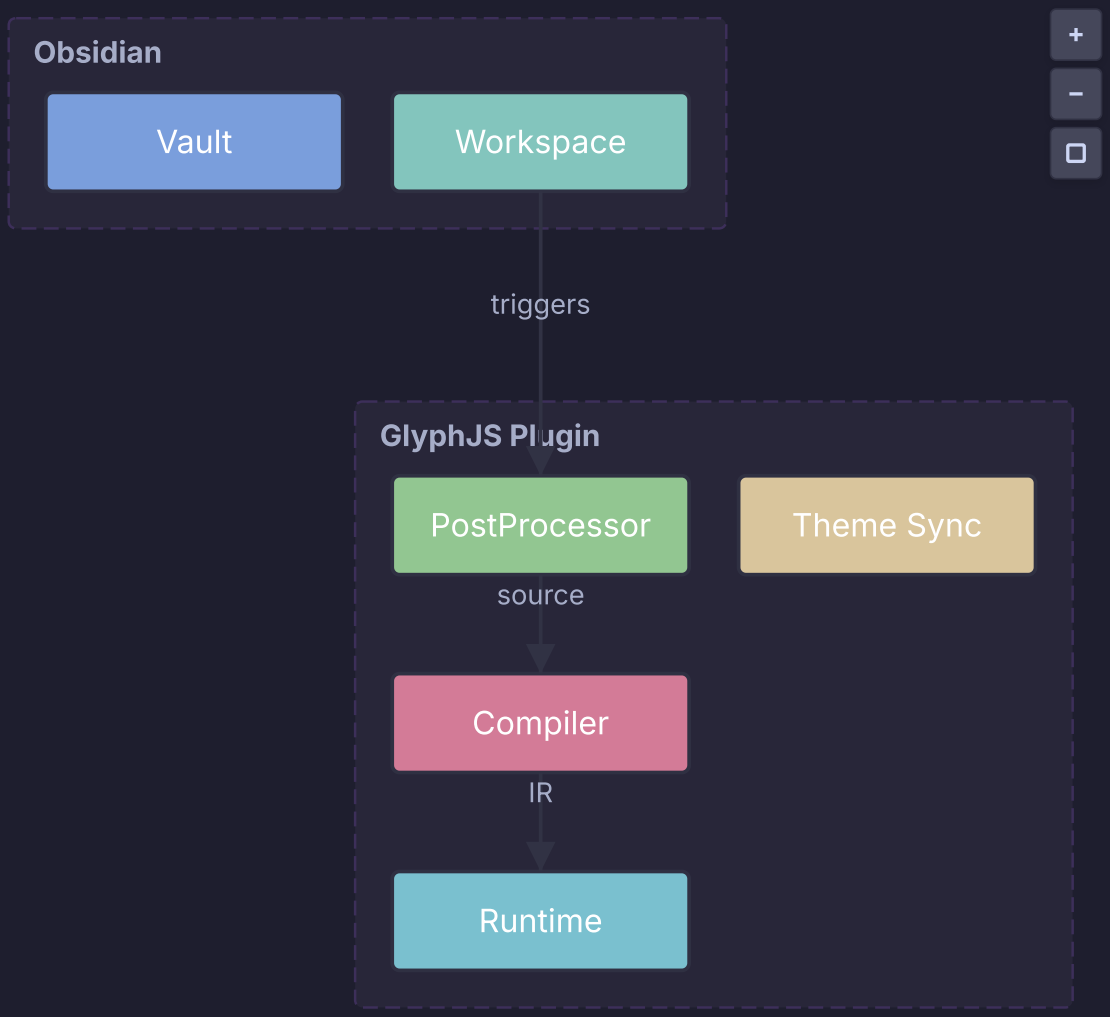
# Graph



## Relation



## Architecture




## Accordion

- Why not registerMarkdownCodeBlockProcessor?
- How does theme sync work?
- What about mobile?

# Comparison

### Integration approach

| Feature | Bundled packages | CLI bridge |
|---|---|---|
| Works offline | ✓ | ✗ |
| Full interactivity | ✓ | ✗ |
| Community plugin eligible | ✓ | ✗ |
| Always latest version | ✗ | ✓ |

# Tabs

**Overview** | Pipeline | Theme

The GlyphJS Obsidian plugin renders interactive components natively in reading view. Uses `registerMarkdownPostProcessor` to scan rendered HTML for `ui:*` code blocks. Each block is compiled independently and mounted with React via `MarkdownRenderChild`.

# Card

**@glyphjs/compiler**

**@glyphjs/runtime**

**@glyphjs/components**

# Infographic

| 28 | 19 MB | 3.7 MB | 1.5.0 |
|---|---|---|---|
| COMPONENTS | BUNDLE RAW | BUNDLE GZIP | MIN OBSIDIAN |

# CodeDiff

```diff
-  1  // Old approach - broken with colons
-  2  this.registerMarkdownCodeBlockProcessor(
-  3    "ui:callout",
-  4    (source, el, ctx) => {
-  5      ctx.addChild(new GlyphBlockChild(el, source));
+     1  // New approach - works reliably
+     2  this.registerMarkdownPostProcessor((el, ctx) => {
+     3    const codeBlocks = el.querySelectorAll('pre > code');
+     4    for (const code of codeBlocks) {
+     5      if (!code.className.startsWith('language-ui')) continue;
+     6      // extract type + source, mount React root
   6  7    }
-  7  );
+     8  });
   8  9
```

## FileTree

```
📂 packages
  📂 obsidian-plugin
    📂 src
      T main.ts
      T theme-sync.ts
    📋 manifest.json
    📄 esbuild.config.mjs
    🎨 styles.css
    📋 package.json
```

## Quiz

1. Which Obsidian API correctly handles ui:* language identifiers with colons?

- registerMarkdownCodeBlockProcessor
- registerMarkdownPostProcessor
- registerEditorExtension

2. What CM6 primitive is required for line-spanning widget decorations?

- ViewPlugin
- StateField
- EditorView.decorations

## Kanban

| TO DO 3 | IN PROGRESS 1 | DONE 2 |
| --- | --- | --- |
| Live preview (Phase 2) | Theme sync | Reading view |
| Mobile support | | All 28 components |
| Settings panel | | |

## Poll

Which GlyphJS component would you use most in Obsidian?

- Table
- Chart
- Callout
- Graph
- Kanban
- Architecture

Vote

## Rating

Ease of setup

★ ★ ★ ★ ★

Documentation quality

★ ★ ★ ★ ★

Rendering performance

★ ★ ★ ★ ★

Theme integration

★ ★ ★ ★ ★

## Ranker

1. Live preview
2. Obsidian theme sync
3. Mobile support
4. PDF export
5. Dataview interop

## Slider

Max bundle size (MB) **0.5**

0.5 — 10

Min Obsidian version (×10) **10**

10 — 20

## Matrix

| | Reading View | Live Preview | Mobile | Total |
|---|---|---|---|---|
| Phase 1 | 0 | 0 | 0 | **0** |
| Phase 2 | 0 | 0 | 0 | **0** |
| Phase 3 | 0 | 0 | 0 | **0** |

## Form

**Plugin ID**

glyphjs

**Description**

Render interactive GlyphJS components in Obsidian

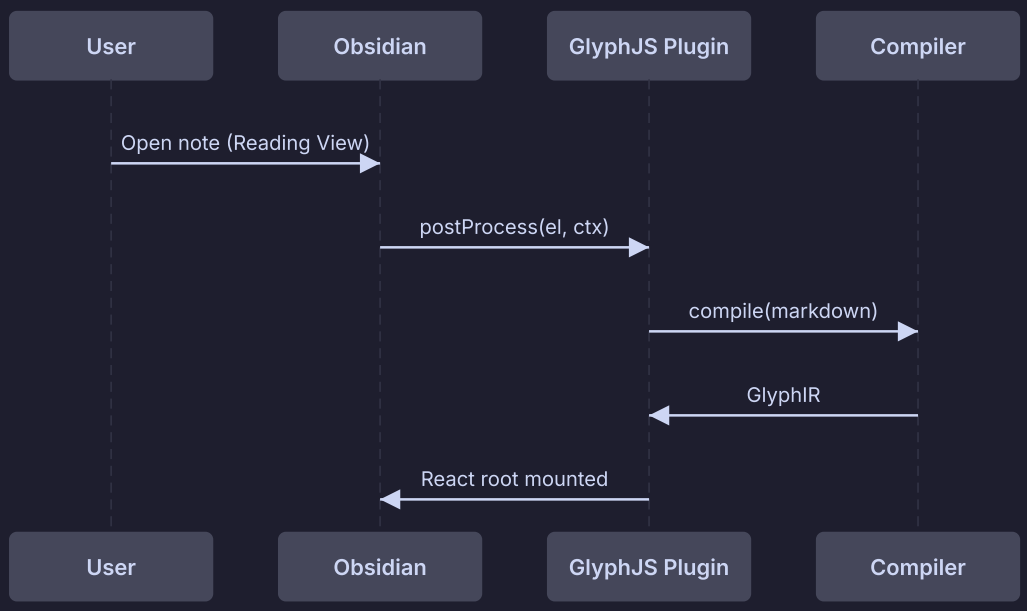
**Current phase**

Phase 1 — Reading view

☐ Mobile support

**Submit**

## Sequence



## MindMap

## Equation

$$E = mc^2$$

E = mc2

Mass-energy equivalence

# Annotate

The plugin uses registerMarkdownPostProcessor to scan rendered HTML for ui:* code blocks, then mounts React roots via MarkdownRenderChild for proper lifecycle management.

**ANNOTATIONS (0)**

Select text to add annotations.